Operational Analytics

Operational analytitcs

Operational analytics is the process of using data analysis to improve efficiency and streamline everyday operations.

A subset of data / business analytics, operational analytics leverages on business intelligence, data mining, artificial intelligence, and machine learning.

1.Faster Decision-Making

Traditionally, businesses make reactive adjustments to their operations based on a quarterly or annual data review.

Businesses that can analyze and react to customer data in real time can make faster adjustments to processes and increase profitability and reduce waste.

2. Faults and failures fixing

They can detect problems and inefficiencies quickly and respond to them rapidly.

3. Increased Productivity

They can see the inefficiencies that exist in their workflows and accordingly change their processes to streamline operations.

Knowledge

ἐπιστήμη		*Epistēmē*
(γνῶσις)		*(gnōsis)*
τέχνη		*Techne*
φρόνησῐς		*Phrónēsis*

Aristotle, The Nicomachean Ethics

Economics (economy)

Οἶκος («household", «family goods") + νόμος ("norm", "law"): studies both … how to make choices on allocating scarce resources to best satisfy wants and needs …

~~Macroeconomics~~

Microeconomics

analyzes basic elements in the economy, including individual agents, … operating with *scarce means*

wikipedia

Normative economics

Economics is concerned with explaining and forecasting observed phenomena.

(micro)economics answers questions both of positive and of normative nature.

- *positive economics* describes "what is", it deals with explanation and forecast,
- *normative economics* advocates *"what ought to be"*, goes beyond explanation and forecast to deal with problems of the kind "what would it be best to do?".

always wikipedia

Solving problems

- Computer science was born (ca. during WW2) to *help solve problems*.
- The first ones, the biggest ones, the most deeply felt were optimization problems.

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>
> *EULER*

- A (computer science) method to solve a problem is an algorithm.
- Solutions are computed for *future* instances of the problem

Data Analytics

Data analytics is the science of analyzing raw data in order to derive conclusions, to optimize processes and increase the overall efficiency of a business or system.

Data analytics is a broad term that encompasses many diverse types of data analysis.

Any type of information can be subjected to data analytics techniques to get insight that can be used to improve processes.

Types of Data Analytics

Data analytics is often divided into *four basic areas*:

- Descriptive analytics describes what has happened over a given period of time. Are sales stronger this month than last?
- Diagnostic analytics focuses on why something happened. This involves a bit of hypothesizing. Did the weather affect beer sales? Did that latest marketing campaign impact sales?
- Predictive analytics forecasts what is likely going to happen in the near term. What happens to sales if we have a hot summer?
- Prescriptive analytics suggests a course of action. If the likelihood of a hot summer, we should add an evening shift to the brewery and rent an additional tank to increase output.

DA process

Data Analytics involves applying an algorithmic process to derive insights. For example, running through a number of data sets to look for meaningful correlations between each other.

Data analysts work with data throughout the data pipeline: mining data, cleaning data, applying analysis techniques, designing programs and databases to manage data.

It is used in a number of industries to allow the organizations and companies to make better decisions as well as to verify or disprove existing theories or models.

The focus of Data Analytics lies in inference: deriving conclusions that are solely based on what the researcher already knows.

EDA and CDA

At a high level, data analytics methodologies include:

- exploratory data analysis (EDA), which aims to find patterns and relationships in data (i.e., make hypothesis),
- confirmatory data analysis (CDA), which applies statistical techniques to determine whether hypotheses about a data set are true or false (i.e., choose an hypothesis).

Relatives: Data Science

Data Science: comprises of everything related to data (unstructured and structured): cleansing, preparation, and analysis.

Data Science is the combination of statistics, mathematics, programming, problem-solving, capturing data in ingenious ways, ability to look at things differently, and activities of cleansing, preparing and aligning the data.

It is the umbrella of techniques used when trying to extract insights and information from data.

Relatives: Big Data

Big Data: "*Big data is high-volume, and high-velocity and/or high-variety information assets that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision making, and process automation*" (Gartner)

Big Data refers to huge volumes of data, unstructured and structured, that *cannot be processed effectively with traditional applications*.

The processing of Big Data begins with the raw data that isn't aggregated and is most often *impossible to store in the memory of a single computer*.

Relatives: Business Analytics

Basically, synonyms.

Data analytics uses algorithms, statistics and big data to identify business questions such as patterns and correlations.
A data analyst collects, processes and analyzes how available data can be used to discover insights that can help businesses improve efficiency.

Business analytics is focused on using the same data tools as data analysis to determine business decisions and implement practical changes within an organization.
More *skewed towards people and processes rather than programming*, statistics and data processing to effectively leverage the data available

Relatives: Business Intelligence

"the difference between Business Intelligence and Data Analytics lies in the fact that Business Intelligence helps in making business decisions based on past results while data analytics helps in making predictions that are going to help in the future."

BI deals with analytics and reporting tools, which helps in determining trends using historical data, It helps individuals in making queries on past data.

Business Analytics helps in determining future trends and rewarding policies.

An archetypical application

Predictive maintenance.

Predictive maintenance is a technique to predict the future failure point of a machine component, so that it is possible to schedule its replacement before failure at minimum generalized cost (min maintenance resources, max device uptime, max process quality).

This technique is based on the measurement of parameters that show a predictable connection with the component life cycle. Examples of such parameters are: vibration, noise, temperature, pressure, oil levels, electrical currents, corrosion, …

PdM optimization

Tries to schedule maintenance before equipment fails but not before needed, avoiding costs associated with repair and production loss.

Difficulties:

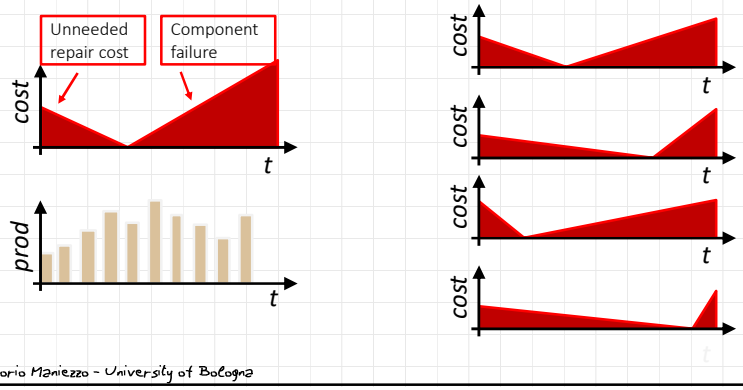
Dependencies: if planned production is highest when maintenance cost is minimum, it could be better to schedule early repairs to avoid peak production loss

Dependencies: it may be possible to group interventions to minimize downtime

Limited resources: if a day is optimal for several components, but less maintenance crews are available

PdM optimization

Optimality for a set of components, subject to limited resources (e.g. maintenance crew availability), other constraints and dependencies (production plan and repair costs), and optimization metrics (min total cost, max customer satisfaction, meeting planned production, min late maintenance).



A logistics application

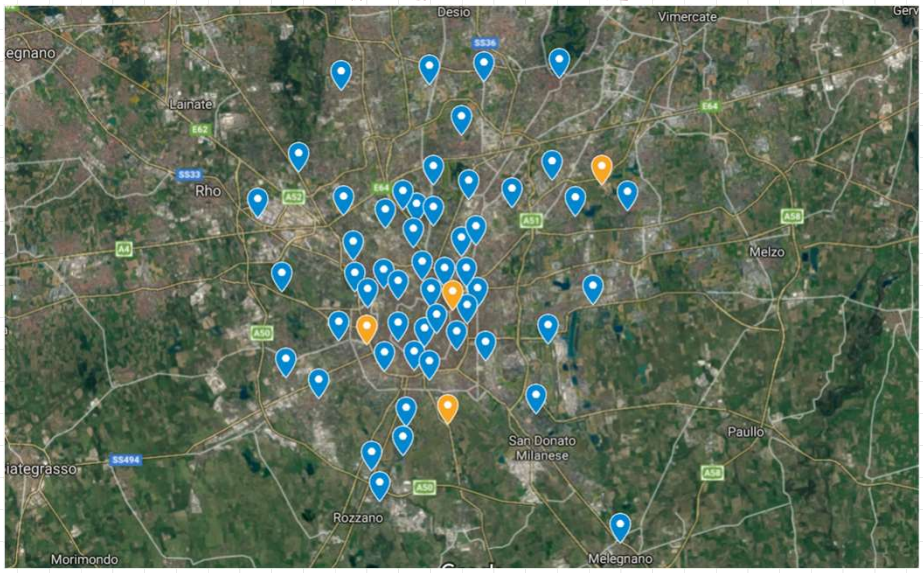
A company has several retail points (supermarkets, SM) in a town, which are resupplied from logistic centers (LC).

Each SM is assigned to one single LC for each merchandise sector (fresh food, canned food, stationery, drinks, etc.).

Different LCs have different, partially flexible warehouse capacities and warehousing costs.

Assignments and capacities are revised each trimester on the basis of expected SM orders.

A logistics application



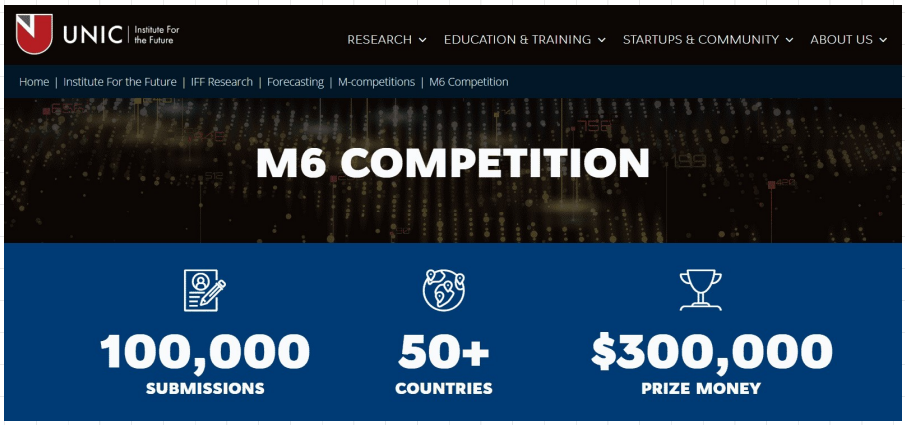
The M competitions

Beginning in 1982, S.Makridakis has led groups of researchers from all over the world in conducting comparisons of various forecasting methods. Data from the M-Competitions have been used by hundreds of researchers (https://en.wikipedia.org/wiki/Makridakis_Competitions).

Datasets (see https://forecasters.org/resources/time-series-data/):

- M(1): https://forecasters.org/resources/time-series-data/m-competition/
- M2: https://forecasters.org/resources/time-series-data/m2-competition/
- M3: https://forecasters.org/resources/time-series-data/m3-competition/
- M4: https://github.com/Mcompetitions/M4-methods/tree/master/Dataset
- M5: https://www.kaggle.com/competitions/m5-forecasting-accuracy/overview
- M6 consisted of online data

M6



https://www.unic.ac.cy/iff/research/forecasting/m-competitions/m6/